Biological Conservation 311 (2025) 111441



Contents lists available at ScienceDirect

Biological Conservation

journal homepage: www.elsevier.com/locate/biocon





Nocturnal flight call monitoring reveals in-flight behavioral alteration by avian migrants in response to artificial light at night☆

Dylan M. Osterhaus [a,b,*], Kelley C. Boland [a], Abigail J. Lawson [c], Kyle G. Horton [d], Benjamin M. Van Doren [e], Patricia L. Cutler [f], Timothy F. Wright [b], Martha J. Desmond [a]

[a] Department of Fish, Wildlife, and Conservation Ecology, New Mexico State University, Las Cruces, NM, USA
[b] Department of Biology, New Mexico State University, Las Cruces, NM, USA
[c] U.S. Geological Survey, New Mexico Cooperative Fish and Wildlife Research Unit, USA
[d] Department of Fish, Wildlife, and Conservation Biology, Colorado State University, Fort Collins, CO, USA
[e] Department of Natural Resources and Environmental Sciences, University of Illinois Urbana-Champaign, Urbana, IL, USA
[f] Environmental Division, U.S. Army Garrison, White Sands Missile Range, NM, USA

ARTICLE INFO

Keywords:
Artificial light at night
Bioacoustics
Bird migration
Light pollution
Migration behavior
Nocturnal flight call

ABSTRACT

The world in which birds evolved to migrate has been drastically altered in the Anthropocene by artificial light. Sources of light such as urban centers or bright upward-facing lights attract migrants, altering their behavior, especially during inclement weather, often leading to mortality. Seemingly less extreme sources, such as pole-mounted floodlighting, ubiquitous throughout much of the world, have received comparatively less study, and migrant responses to such sources are poorly understood. We studied migrant behavior in relation to light at White Sands Missile Range (New Mexico, USA) by recording nocturnal flight calls at sites with and without lights during non-inclement weather. We collected 103,424 h of recordings and detected 2,851,863 calls over three fall migration seasons. We assessed how temporal, weather, and lighting variables explain variability in call rates between light and dark sites, and examined how different taxonomic groups behave in relation to light. Contrary to predictions, call rates were higher at dark sites than at light sites, and this difference was strongest early in the migration season. We found illuminated sites with a greater proportion of shielded lights, or with lights of higher dominant wavelengths (warmer color temperatures), had higher call rates (closely resembling dark sites) than other light sites, indicating that these factors may reduce impact to migrants. Our taxonomic analyses revealed consistent differences in call rate between light and dark sites for warblers, but no difference for most sparrows. Our findings indicate that lights alter behavior, but the use of "bird-friendly" lighting strategies may reduce this impact.

1. Introduction

Avian migration involves the seasonal movements of billions of birds, which have evolved a myriad of physiological, sensory, and cognitive adaptations to complete these journeys (Berthold et al., 2013). The world in which these adaptations evolved has been profoundly altered in the Anthropocene. Humans have altered habitats at large scales (Gaston et al., 2003), erected physical obstacles in migratory paths (Loss et al., 2015), and created phenological mismatches between migrants and the resources they depend on through climate change (Jones and Cresswell, 2010; Saino et al., 2011). As a result, migratory bird populations have experienced declines worldwide (Bairlein, 2016; Rosenberg et al., 2019).

The proliferation of artificial light at night (hereafter, ALAN) is one of the most pervasive forms of environmental alteration experienced by nocturnal avian migrants (Cabrera-Cruz et al., 2018; Horton et al., 2019). Having evolved to migrate in a landscape with no artificial light, nocturnal migrants now must navigate through a night where 80 % of the world exists under a light-polluted sky (Falchi et al., 2016). ALAN alters the natural migration behavior of nocturnal migrants at both local (Burt et al., 2023b; Rebke et al., 2019; Van Doren et al., 2017) and regional scales (Cabrera-Cruz et al., 2018; Horton et al., 2023; McLaren

☆ This article is part of a special issue entitled: 'Dark night conservation' published in Biological Conservation.

* Corresponding author at: 2980 South Espina, Knox Hall 132, Las Cruces, NM 88003, USA.
E-mail address: dylano@nmsu.edu (D.M. Osterhaus).

https://doi.org/10.1016/j.biocon.2025.111441
Received 7 March 2025; Received in revised form 17 July 2025; Accepted 13 August 2025
Available online 29 August 2025
0006-3207/© 2025 The Authors. Published by Elsevier Ltd. This is an open access article under the CC BY-NC license (http://creativecommons.org/licenses/by-nc/4.0/).

et al., 2018), with juveniles seeming to alter behavior more strongly than adults (Beacham, 1982; Gauthreaux, 1982; Williams, 1978). Locally, migrants may exhibit positive phototaxis and deviate from normal migratory flight to circle around lights, especially under poor weather conditions or when lights are extremely bright, installed high above ground-level, and are unshielded or upward-facing (Gauthreaux Jr. et al., 2006; Van Doren et al., 2017). The presence of ALAN is correlated with increased avian stopover densities at a landscape scale, demonstrating that the impacts of ALAN extend into daylight hours by altering stopover habitat selection (Cabrera-Cruz et al., 2020; Horton et al., 2023; McLaren et al., 2018). The attraction of migrants to ALAN has been implicated as an important factor contributing to the injury and death of migrants via collisions with infrastructure (Loss et al., 2019; Tan et al., 2024; Van Doren et al., 2021). Attraction to ALAN also may indirectly impact migrants via increased energetic costs if birds deviate from preferred migratory paths or land in sub-optimal stopover habitat (Cabrera-Cruz et al., 2019; Lindstrom, 2003; Nemes et al., 2023). Given the intense energetic demands of migration, excess energy expenditure may be detrimental to migrants and could lead to reduced fitness or survival.

Previous studies have examined how migrants respond to ALAN at large scales (Horton et al., 2023; McLaren et al., 2018) or to single extremely high-intensity sources of ALAN (Van Doren et al., 2017), and others have noted behavioral change in poor weather (Zhao et al., 2014). Fewer studies have examined behavior at sources of ALAN more typical of the residential, commercial, or peri-urban settings that are ubiquitous throughout the populated world. These few studies have found variable behavioral responses including changes in flight altitude, both toward and away from lights, as well as altered nocturnal flight call behavior (Cabrera-Cruz et al., 2021; Gillings and Scott, 2021; Watson et al., 2016). Additionally, little attention has focused on species-specific or age-specific in-flight responses to ALAN (Winger et al., 2019). Lastly, few studies have assessed how avian migrant behavior may vary in relation to different lighting metrics (dominant wavelength, illuminance, etc.), or different installments (shielded versus unshielded lights), or how these variables may interact and explain behavioral alteration (Rebke et al., 2019; Zhao et al., 2020, 2014). Currently, there is limited resolution in our understanding of how nocturnal migrants behave in relation to ALAN that is not of extremely high intensity or during non-inclement weather. An improved understanding of avian behavior in response to the sources of ALAN typical of residential, commercial, and peri-urban settings may facilitate evidence-based lighting improvements to reduce adverse impacts to birds.

To improve understanding of how migrants interact with ALAN, we studied avian migration using nocturnal flight call monitoring at an array of recording locations. Our study was conducted on White Sands Missile Range (hereafter, WSMR) in southern New Mexico, USA, a location that has experienced previous instances of nocturnal migrant mortality, including one that was associated with five unshielded light-emitting diode (LED) floodlights during inclement weather in the fall of 2015 (Boland, 2024; Roper, 2015). Approximately half of the recording locations for the current study included different types of lighting installations (hereafter, light sites), while the remainder of recording locations did not (dark sites). At each location, we placed an autonomous recording unit (ARU) programmed to record nocturnal flight calls throughout each night of the fall migration season (Osterhaus et al., 2025b). We used nocturnal flight call rate (hereafter, call rate; calls hr^{-1}) to assess how migrants behaved at light versus dark sites. Call rates measured using ARUs have been demonstrated to correlate with the density of migrants aloft in active migration and are a measure of migration activity (Horton et al., 2015; Osterhaus et al., 2024). Any detectable difference in call rate between light and dark sites would be indicative of behavioral alteration and deviation from normal migratory behavior as we considered dark sites to be representative of "natural" migration conditions. In addition to nocturnal flight call monitoring, we measured a suite of variables relating to the type of ALAN present at each site (number of lights, illuminance, dominant wavelength, etc.), weather conditions (moon illumination, cloud cover), and temporal variables (time of night, time of migration season), and generated multiple metrics of migration activity and behavior from weather radar data. We aimed to describe specific lighting factors that may be disruptive to migrants, identify weather conditions that may amplify disruption caused by ALAN, and determine whether ALAN is particularly disruptive during select temporal periods or for certain taxonomic groups of migrants.

Based on existing research, we hypothesized that migrants would exhibit positive phototaxis toward sources of ALAN, drawing more migrants within the sampling area of ARUs at light sites, leading to an increase in recorded call rates at light sites relative to dark sites. Therefore, we predicted that call rates would be higher at light sites than dark sites (Gillings and Scott, 2021; Watson et al., 2016). Also, we expected increased call rates with increasing number of individual lights, increasing brightness of lights, and decreasing wavelengths of light (i.e., cooler-colored lights) (Zhao et al., 2020, 2014). Lastly, we predicted that differences in call rate between light and dark sites would not vary across the migration season or between taxa, as previous studies have not detected a temporal or taxonomic trend in attraction, suggesting responses are similar across time and taxa. Overall, our findings serve to increase our understanding of how migrants interact with ALAN and provide conservation practitioners with guidance to limit potential effects of ALAN on nocturnal avian migrants.

2. Methods

2.1. Study area

We monitored nocturnal migration using ARUs on WSMR (32.4020°N, 106.3329°W) in the Tularosa Valley of south-central New Mexico, USA (Fig. 1). The Tularosa Valley is located within the Chihuahuan Desert ecoregion at approximately 1220 m elevation and is sparsely populated. In this region, extensive sources of ALAN typical of small urban centers are primarily confined to the cities of Alamogordo, New Mexico (~80 km NE), neighboring Holloman Air Force Base (~65 km NE), and El Paso, Texas (~70 km S) (Fig. 1). At WSMR, ALAN is generally concentrated at the WSMR Post Area (hereafter, Main Post; typical of a small urban center) and at dispersed test sites within the Tularosa Valley, primarily along an east-west transect (Fig. 1). Lighting applications include parking and street lighting, safety and security lighting at low-rise office complexes and residential areas, and task and security lighting at test facilities. Type and intensity of lighting are variable, and range from a handful of low-intensity high-pressure sodium (HPS) sources to over 100 high-intensity LED sources at high-security sites. In 2019, WSMR began the large-scale replacement of HPS floodlights with commercial high-intensity LEDs to meet energy conservation goals. Most floodlights are controlled by photocells rather than dimmers, timers, or motion sensors and remain on all night.

2.2. Sampling sites and acoustic data collection

We collected recordings at 50, 55, and 56 sites during fall migration of 2021, 2022, and 2023, respectively. At all sites, a single ARU (Wildlife Acoustics Song Meter Micro; 44,100 Hz sampling frequency; 12 dB gain) with an omnidirectional microphone was affixed to a metal t-post or similar structure at a height of ~1.5 m, oriented to manufacturer specifications. The maximum sampling range of ARUs for nocturnal flight calls has been estimated to be from 200 to 800 m above ground level, depending on the species calling (Evans and Mellinger, 1999; Evans and Rosenberg, 2000). Recorders were programmed to record continuously from 1.5 h after sunset to 1.5 h before sunrise from Sept 1 – Nov 3 in 2021, Sept 1 – Nov 9 in 2022, and Aug 17 – Nov 12 in 2023. We chose this nightly sampling timespan to avoid recording the calls of birds on the ground and not in active migration (Osterhaus et al., 2024). We

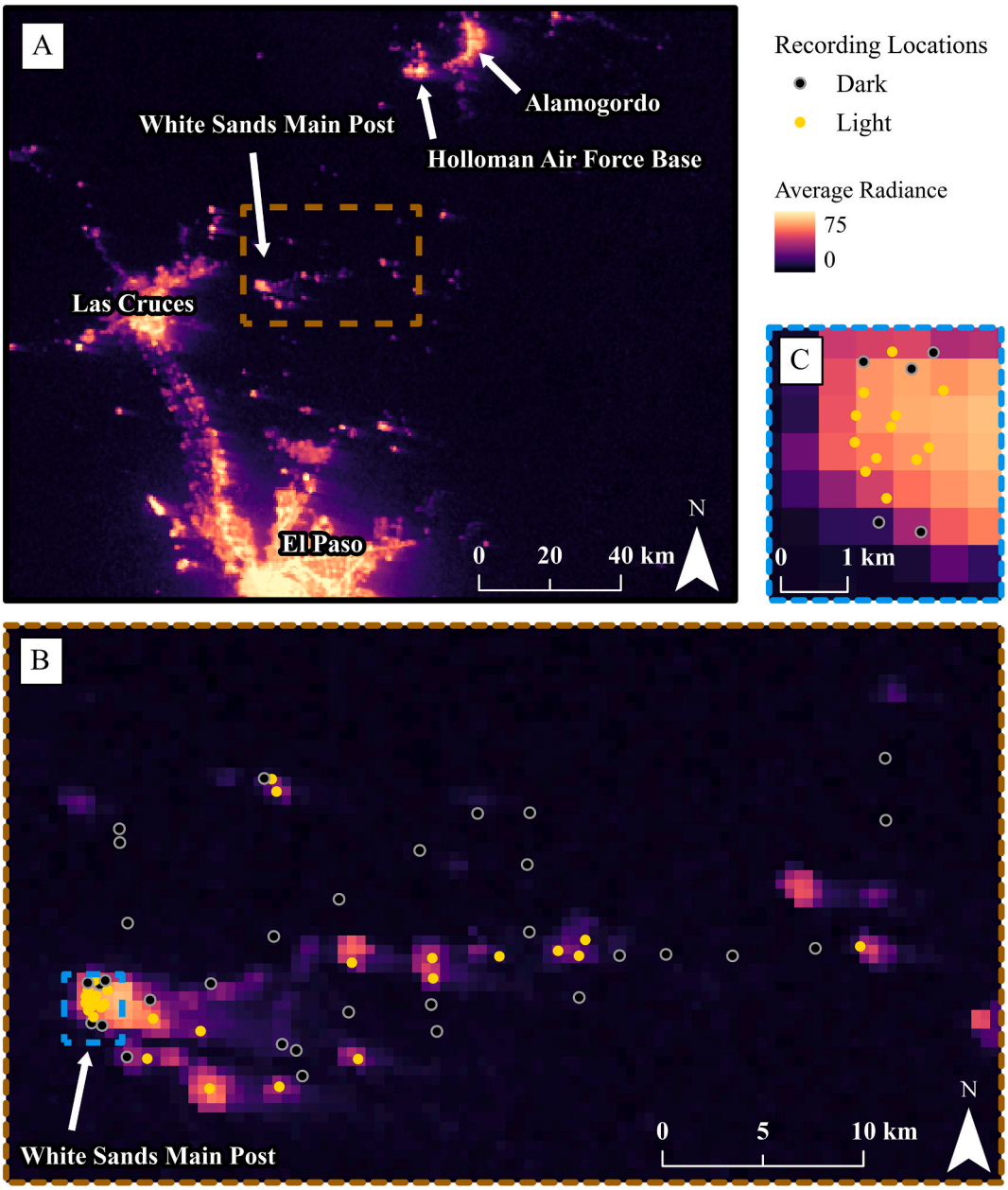


Fig. 1. Map of the study area where (A) depicts presence of sources of artificial light at night at the regional scale in 2024 (south-central New Mexico and west Texas, USA) surrounding the area where acoustic recordings were collected (outlined in brown dashed line)*, (B) spatial layout of acoustic recording locations within White Sands Missile Range in south-central New Mexico and White Sands Main Post (outlined in blue dashed line), (C) spatial layout of acoustic recording locations within White Sands Main Post area. The basemap for each panel depicts average radiance data (a measure of the brightness/intensity of artificial light at night) at ~500 m resolution, obtained from the NASA/NOAA Visible Infrared Imaging Radiometer Suite (VIIRS). *Since 2016, White Sands Missile Range has begun transitioning to LED lighting which is not accurately quantified by VIIRS data (Hung et al., 2021). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

classified sites by their lighting status and labelled sites as "dark" if there were no direct sources of ALAN present within 200 m of the ARU (26 in 2021, 29 in 2022, 27 in 2023) and "light" if there was a direct source of ALAN (24 in 2021, 26 in 2022, 29 in 2023). Mean illuminance was 30.44 lx at light sites and 0.06 lx at dark sites (Supplementary Fig. S1). ARUs at light sites were placed within 5 m of an ALAN source.

2.3. Acoustic data processing

We processed recordings and extracted nocturnal flight call data using Nighthawk (Van Doren et al., 2024), a deep learning model trained to detect and classify nocturnal flight calls in acoustic recordings. To maximize detection rates, the core Nighthawk model (freely available at https://github.com/bmvandoren/Nighthawk/) can be fine-tuned using data from the focal study area. The version of Nighthawk used for our study was fine-tuned using ~700 h of recordings collected within our study area, annotated to the order level by D. Osterhaus (~500 h and ~200 h for 2021 and 2022, respectively).

Using the fine-tuned model, we created two datasets of call rates. For Dataset I, which we used to assess overall avian migrant responses to ALAN, we retained all detections with classification probability values > 0.5 and retained only calls from Order Passeriformes as classified by Nighthawk (representing 98 %, 97 % and 99 % of all calls identified to order in 2021, 2022, and 2023, respectively). Output from this fine-

tuned version of Nighthawk at these classification thresholds is an accurate index of call rates for Order Passeriformes for data from 2021 and 2022 within our study system (Osterhaus et al., 2024). We confirmed that the fine-tuned version of Nighthawk performed well on data from 2023 by randomly selecting 50 15-min segments of recordings from 2023, manually counting nocturnal flight calls within these segments, and comparing manual counts to those generated by Nighthawk, finding a strong correlation ($r = 0.97$) between manual and automated counts (Supplementary Fig. S2).

To create Dataset II, used to assess taxon-specific responses to ALAN, we retained all detections with classification probability values >0.5 and retained only calls from the families Parulidae and Passerellidae as classified by the fine-tuned Nighthawk model (representing 95 %, 92 % and 97 % of all calls identified to family in 2021, 2022, and 2023, respectively). Additionally, we retained all detections with probability values >0.8 at the species level for species with >5000 detections, which included Chipping Sparrow (*Spizella passerina*), Grasshopper Sparrow (*Ammodramus savannarum*), Savannah Sparrow (*Passerculus sandwichensis*), Vesper Sparrow (*Pooecetes gramineus*), Wilson's Warbler (*Cardellina pusilla*), and Yellow-rumped Warbler (*Setophaga coronata*). Lastly, Nighthawk also classifies calls to similar-sounding species groups (hereafter, groups) that produce vocalizations too similar to distinguish at the species level. We retained all detections with probability values >0.8 for these groups, and retained all groups with >5000 detections that did not include a species mentioned above. These groups (as defined by Nighthawk) were CCBRS (Clay-colored/Brewer's Sparrow), DBUP (double-banded up frequency calls), SFHS (single-banded frequency modulated up-arching calls), SWLI (Swamp Sparrow/Lincoln's Sparrow), and ZEEP (short high-frequency buzzy calls; descriptions of species included within groups can be found in Supplementary Table S1). To ensure that Nighthawk was performing well in identifying calls at the species and group levels, we randomly selected 200 calls from each species or group listed above and manually identified these calls. We found that Nighthawk had >94 % accuracy in identifying calls for each species or group that we analyzed (Supplementary Table S1).

For each night of data collection, we divided the night into ten equal temporal segments (deciles) and calculated call rates (calls hr^{-1}) for each site during each decile of each night. Decile 1 began when recordings began (1.5 h after sunset) and decile 10 ended when recordings ended (1.5 h before sunrise). Throughout the recording season, decile length ranged from 0.79 h to 1.06 h, corresponding with seasonal changes in night length.

2.4. Weather data and temporal variables

To account for other potential drivers of variability in call rates, we used weather covariates of wind (tailwind and crosswind), cloud cover, and moon illumination, and temporal covariates of decile, night, and year. Wind data were provided by WSMR Meteorology Branch and were collected at a height of 10 m at a location near the center of our acoustic recording locations. We averaged wind speed and direction within each decile of data collection and followed the methods of Osterhaus et al. (2024) to calculate tailwind assistance (m s^{-1}, hereafter tailwind), and absolute crosswind (m s^{-1}, hereafter crosswind), assuming migrant flight direction was due south (180°). Equations for tailwind and crosswind can be found in Supplementary Table S2. Cloud cover data were accessed from the Holloman Air Force Base weather station, located ~52 km from the center of our recording locations, and classified on a scale of 1 (no cloud cover) to 5 (complete cloud cover) and averaged within each decile for each night. Moon illumination data, corresponding to the proportion of the moon illuminated above the horizon, were collected for the location corresponding with the center of our recording locations for all deciles using the getMoonIllumination() and getMoonTimes() functions from suncalc in R (Thieurmel and Elmarhraoui, 2022). The night covariate corresponds to the ordinal date at sunset on the night for which the data were collected (e.g., night 244 corresponded to data collected between sunset on September 1st to sunrise on September 2nd in a non-leap year). Finally, we assigned a categorical variable of year for the year in which data were collected (2021, 2022, or 2023).

2.5. Radar variables

Given their known relationships with call rates, we sampled average migrant height and bird-migration activity via reflectivity (vertically integrated reflectivity, hereafter radar reflectivity) using unfiltered (level II) Next Generation Weather Radar data from the KHDX radar site (33.0770°N, 106.1346°W), located ~75 km from the center of our recording locations (Osterhaus et al., 2024). We followed the methods of Osterhaus et al. (2024), and used the bioRad package in R (Dokter et al., 2019) to download and process radar data. We downloaded radar scans for the time period matching acoustic data collection and created vertical profiles of reflectivity for each scan at 100-m intervals from 100 to 3000 m above ground level using the 5 lowest elevation scans (0.5° to 4.5°) within a range of 5 to 37.5 km from the KHDX radar site. We integrated data across vertical profiles using the integrate_profile(). Prior to integration, we removed precipitation and retained biological targets from the unfiltered data by eliminating data with correlation coefficient > 0.95 or reflectivity factor > 35, as these data are likely contaminated by precipitation or non-avian targets. Additionally, we removed all scans in which biological targets were moving in the direction opposite of the expected migratory direction for our area (i.e., average azimuths between 90° and 270° were retained). From the processed data, we extracted an estimate of radar reflectivity (in cm^2 km^{-2}) and migrant flight height (in m above sea level) for each scan. We then calculated average radar reflectivity and average migrant height for each decile of each night that acoustic data were collected.

2.6. Lighting variables

From each ARU, a 200 m buffer was used to define the area in which we measured lighting variables because this describes the estimated effective sampling range of ARUs for nocturnal flight calls (Evans and Mellinger, 1999; Evans and Rosenberg, 2000). At light sites, we counted all lights within the 200 m buffer and calculated the proportion of lights as shielded by dividing the number of shielded lights by the total number of lights at a site. A light source was considered "unshielded" if it allowed light to be projected vertically above a horizontal plane at the top of the light fixture. Lastly, we used a UPRtek MK350s Premium All-Purpose Light Meter (UPRtek, Zhunan, Taiwan) to measure dominant wavelength and correlated color temperature (CCT) from the ground at each light source within the 200 m buffer. These measurements were then averaged across all light sources within the site to generate a single value for each variable for that site. At both light and dark sites, we measured illuminance from the ground at the ARU using a ExTech EA30 Easy View Wide Range Light Meter (ExTech Industrial Electronics, Knoxville, TN, USA). Additionally, at all sites we counted all lights within 500 m of the ARU location to understand how lighting at larger scales may alter behavior. A description of each variable and their definitions can be found in Supplementary Table S2, and summaries for each of the lighting variables can be found in Supplementary Fig. S1.

2.7. Modeling

To assess variability in call rate in relation to the presence or absence of ALAN and lighting and weather variables, we generated and fitted two generalized linear mixed models with a Gaussian distribution using the glmmTMB() function from the glmmTMB package in R (Brooks et al., 2017; Table 1). For each model, we applied a square-root transformation to call rates from Dataset I and used this as the response variable. We square-root transformed call rate to alleviate non-normality of model residuals when using non-transformed call rate. Both models included

Table 1
Model specifications for each model used to analyze variability in call rates from data collected from southern New Mexico during fall migration of 2021, 2022, and 2023. Interaction terms that were found to be non-significant and were subsequently removed from the model are *italicized*. Square-root transformed call rates was the response variable for each model with Model I using call rates from all sites and Model II using data from light sites only. For the taxonomic model, individual models were fitted for each taxonomic group.

Model	Equation
I	Light Status + Light Status*Decile + Light Status*Night + Light Status*Cloud Cover + Light Status*Moon Illumination + *Light Status*Num. Lights (500 m)* + *Light Status*DTM_km* + *DTM_km*Num. Lights (500 m)* + Num. Lights (500 m) + DTM_km + Cloud Cover + Moon Illumination + Illuminance + Radar Reflectivity + Avg. Migrant Height + Crosswind + Tailwind + Decile + Night + I(Night^2) + I(Night^3) + Year + (1|Site) + (1|Night)
II	Dom. Wavelength + Num. Lights (200 m) + Prop. Shielded + Dom. Wavelength*Num. Lights (200 m) + Dom. Wavelength*Prop. Shielded + *Dom. Wavelength*Cloud Cover* + Dom. Wavelength*Moon Illumination + *Illuminance*Cloud Cover* + Illuminance*Moon Illumination + Illuminance*Prop. Shielded + Illuminance*Num. Lights (200 m) + Num. Lights (200 m)*Prop. Shielded + *Num. Lights (200 m)*Cloud Cover* + *Num. Lights (200 m)*Moon Illumination* + *Prop. Shielded*Cloud Cover* + *Prop. Shielded*Moon Illumination* + Num. Lights (500 m) + DTM_km + Cloud Cover + Moon Illumination + Illuminance + Radar Reflectivity + AvgMigrantHeight + Crosswind + Tailwind + Decile + Night + I(Night^2) + I(Night^3) + Year + (1|Site) + (1|Night)
Taxonomic	LightStatus + Night + Year + I(Night^2) + I(Night^3)

random intercepts for site and night. Each model also included illuminance, number of lights (500 m), distance to Main Post, cloud cover, moon illumination, average migrant height, crosswind, tailwind, decile, radar reflectivity (continuous), night (ordinal date at sunset on the night the data were collected, including quadratic and cubic terms), and year (categorical). We chose to include this suite of variables within each model to account for factors that create variability in call rates that cannot be attributed to the presence of ALAN alone (such as weather or temporal variables), or to account for variation in the lighting environment at large scales across our study area (illuminance, number of lights (500 m), distance to Main Post). Accounting for these variables in each model aids in model fit and the estimation of the additional parameters included in each model.

Model I (Table 1) was designed to investigate how the presence of ALAN (light status) may interact with temporal (decile, night), weather (cloud cover, moon illumination), and large-scale lighting variables (number of lights (500 m), distance to Main Post) to contribute to variability in call rate. Within Model I, we also assessed how the interaction of large-scale lighting variables (number of lights (500 m), distance to Main Post) may contribute to call rate variability. Output from Model I is most informative for conservation strategies focusing on altering the presence of lighting temporally or spatially to protect migrants. Model II (Table 1) used data from lighted sites only and was designed to investigate how lighting variables (dominant wavelength, number of lights (200 m), proportion shielded, illuminance) may interact, or how lighting variables may interact with weather variables (cloud cover, moon illumination) to contribute to variability in call rate. Output from Model II is most informative for developing best management practices related to specific types of lighting installations (e.g., number of lights, shielding, spectrum) to mitigate effects of ALAN on migrants.

Prior to model fitting, we checked for collinear variables and if two variables were found to be highly collinear ($r > 0.7$) we removed one of them. Dominant wavelength and CCT were collinear, and we retained dominant wavelength. All continuous predictors were z-standardized to have a mean of 0 and variance of 1 prior to modeling. We considered variables significant if the 95 % confidence interval for the coefficient estimate for the variable did not overlap zero. We removed all non-significant interactions from each of the original models and present the models with these terms removed. Finally, to ensure spatial independence, we conducted a Moran's I test on the residuals from Model I using the spdep package in R (Bivand et al., 2025), and found no evidence of significant spatial autocorrelation ($p = 0.82$), indicating the model is adequately accounting for spatial effects.

2.8. Taxonomic analysis

We compared call rates within each taxonomic grouping (family, species, and Nighthawk "group") between light and dark sites using Dataset II. For each taxonomic grouping, we fitted generalized linear mixed models with a Gaussian distribution as above, with square-root transformed call rate measured at a nightly level across sites as the response. We included the covariates of lighting status, night (cubic polynomial), and year in the models (Table 1). All analyses were completed and figures generated in R (R Core Team, 2023).

3. Results

Over the three fall migration seasons, we collected 103,424 h of recordings over 218 unique sampling nights (Supplementary Table S3). Within these recordings, Nighthawk detected and identified 2,851,863 nocturnal flight calls to the order level (Supplementary Table S3).

3.1. Modeling results

3.1.1. Model I

Model I identified multiple variables and interactions as significant predictors of call rates (Fig. 2a; for full model output see Supplementary Table S4). The interactions between light status and number of lights (500 m), light status and distance to Main Post, and distance to Main Post and Number of Lights (500 m) were not significant predictors of call rates and were removed from the final model (Table 1). Call rates differed across years, but we observed similar trends in call rates across nights of the recording season (Figs. 2b, 3). Call rates increased with increasing radar reflectivity and decreased with increasing average migrant height and crosswind, whereas tailwind was not a significant predictor (Fig. 2a). Call rates were lowest in deciles closest to sunset and highest in deciles nearest to dawn. Calling increased more rapidly throughout the night at light sites relative to dark sites (Fig. 2c). Call rates were higher at dark sites than light sites early in the recording season, but this difference disappeared later in the season (Fig. 2d). As cloud cover increased, call rates decreased at dark sites but remained consistent at light sites (Fig. 2e). Lastly, as moon illumination increased, call rates decreased at light sites and increased slightly at dark sites (Fig. 2f).

3.1.2. Model II

Model II identified multiple variables and interaction terms as significant predictors of call rates (Fig. 4a, model output in Supplementary Table S5). The interactions between dominant wavelength and cloud cover, illuminance and cloud cover, number of lights (200 m) and cloud cover, number of lights (200 m) and moon illumination, proportion shielded and cloud cover, and proportion shielded and moon illumination were not significant predictors of call rates and were removed from the modeling output reported here (Table 1). Year, night, decile, radar reflectivity, average migrant height, crosswind, and tailwind displayed similar relationships as in Model I (Supplementary Table S5). As number of lights (200 m) increased, call rates increased when dominant wavelength was high and decreased when dominant wavelength was low (Fig. 4b). As proportion shielded increased, call rates increased for sites with high dominant wavelength, but remained consistent or slightly decreased for sites with low dominant wavelength (Fig. 4c). Though statistically significant, the interaction between moon illumination and dominant wavelength had a small effect, and was poorly distinguished

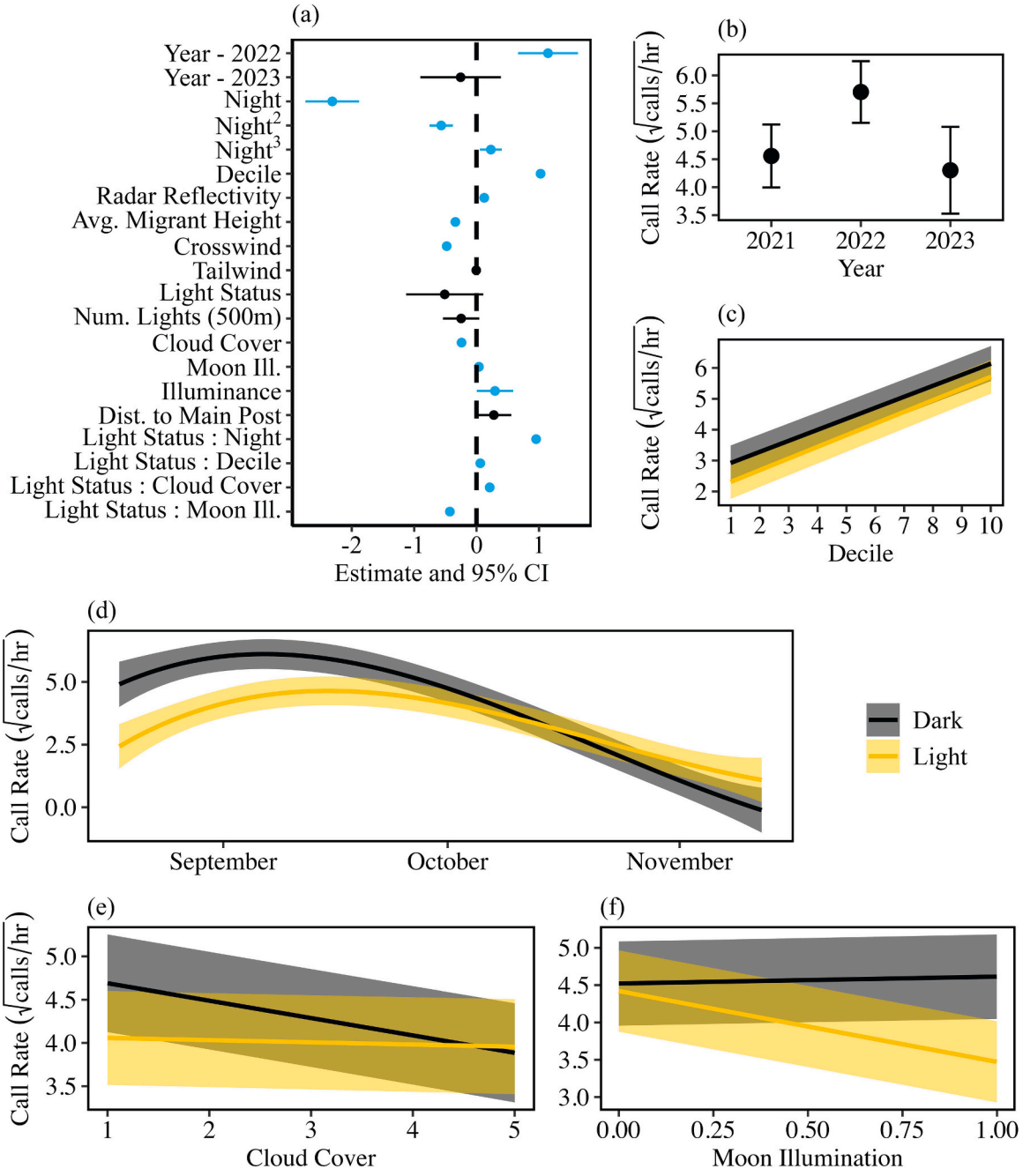


Fig. 2. Output from Model I assessing how call rates vary between light and dark sites for temporal and weather variables for data collected in southern New Mexico during fall migration of 2021, 2022, and 2023. (a) Coefficient estimates and 95 % confidence intervals for each predictor in the model (dots and whiskers in black overlapped zero and are considered non-significant while those in blue do not overlap zero and are considered significant predictors of call rates), and the predicted relationships between call rates and (b) year, and the interactions between lighting status and (c) decile, (d) night of the recording season (from 17 August to 12 November), (e) cloud cover, and (f) moon illumination. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

among the wavelength classes (Fig. 4d). As the number of lights within 200 m of the recording location increased, call rates increased when illuminance was high and decreased when illuminance was low (Fig. 4e). As proportion of shielded lights increased, call rates increased at sites where illuminance was high and decreased at sites with low illuminance (Fig. 4f). While statistically significant, the interaction between moon illumination and illuminance on call rates was relatively weak (Fig. 4g). Lastly, as the number of lights within 200 m of the recording location increased, call rates remained consistent for sites with high proportion shielded and decreased for sites with low proportion shielded (Fig. 4h).

3.2. Taxonomic analysis

The models assessing taxon-specific variation in call rates in response to light identified light status as a significant predictor of call rates for Family Parulidae, the species Chipping Sparrow, Wilson's Warbler, and Yellow-rumped Warbler, and the Nighthawk groups DBUP, SWLI, and ZEEP (Fig. 5a, Supplementary Table S6). Call rates were higher at dark sites than at light sites for each of these groups other than SWLI, which

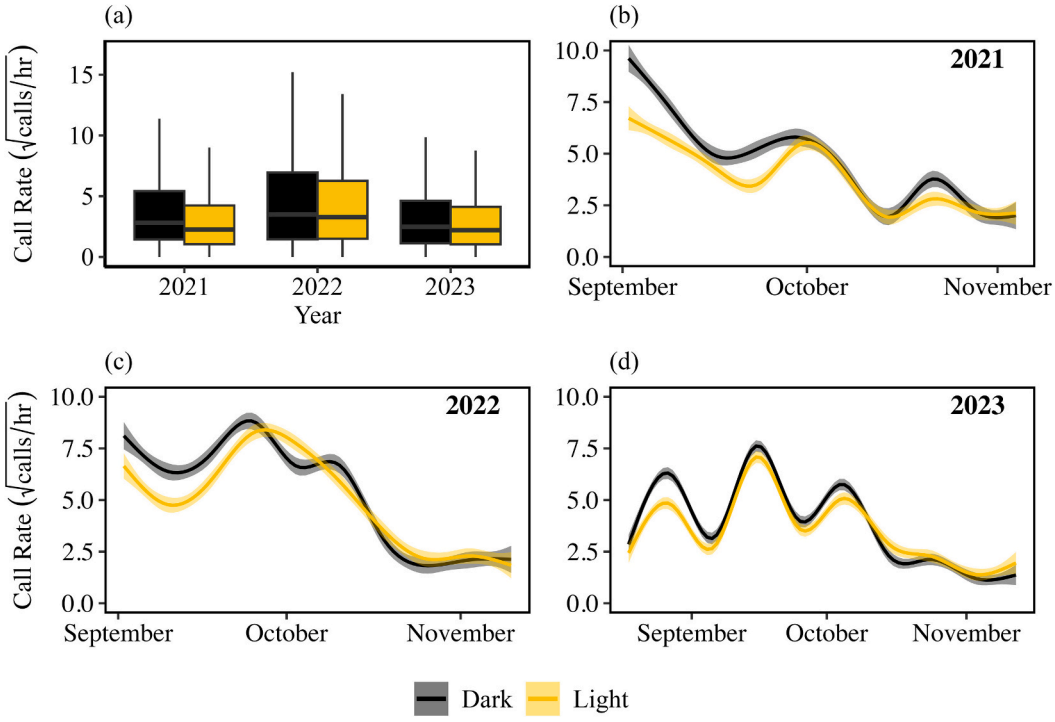


Fig. 3. Comparing call rates between light and dark sites by (a) boxplots for light and dark sites for each recording season and, (b–d) call rates at light and dark sites across nights of each recording season for data collected in southern New Mexico during fall migration of 2021, 2022, and 2023 respectively. Curves presented are generated using the geom_smooth function from ggplot2 with method = "gam" and include 95 % confidence intervals. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

had higher call rates at light sites than dark sites (Fig. 5b).

4. Discussion

Artificial light at night (ALAN) is widespread throughout the world and has been shown to cause behavioral alterations in nocturnal avian migrants at local and regional scales. Our study, comparing avian migrant behavior between dark sites without ALAN and sites with ALAN typical of lighting ubiquitous in residential, commercial, peri-urban, and urban settings, found that birds altered their in-flight behavior in response to ALAN. Contrary to our predictions, we found nocturnal flight call rates to be higher at dark sites than light sites, but only early in the migration season. In addition to this unexpected finding, we identified a suite of variables that are linked to the difference in call rates between light and dark sites, including proportion of lights shielded and the dominant wavelength of the lighting. Also, we found that the effect of light on call rates varied between taxonomic groups, but most showed higher call rates at dark sites. Our findings raise new questions regarding how ALAN may alter the behavior of migrants during non-inclement weather, and our identification of drivers of this alteration reveal potential areas of focus for protecting migrants from ALAN.

4.1. Behavioral alteration in relation to ALAN

Many prior studies and anecdotal accounts detailing avian attraction to ALAN have noted that attraction occurred primarily during nights with weather conditions that were not conducive to migration (strong headwinds, extensive cloud cover, precipitation) (Gauthreaux Jr. et al., 2006), at extremely high-intensity sources of ALAN that were upward facing (Van Doren et al., 2017), or ALAN installed high above the ground (such as on skyscrapers or radio towers; Gauthreaux Jr. et al., 2006; however, see Van Doren et al., 2021). Studies that have found higher call rates at light sites relative to dark sites have either relied on data collected on a single night with no detailed description of weather conditions, which may influence attraction to ALAN (Watson et al., 2016), or focused on a taxonomic group of migrants that pass through our study area in very low numbers (Gillings and Scott, 2021). Our study region has relatively stable weather and few nights with inclement weather conditions. Also, the sky glow associated with our study area is generally not as bright as that of large urban centers, skyscrapers, communication towers, or extremely bright upward facing lights such as spotlights, and the surrounding landscape is primarily dark. Therefore, we believe that our findings are representative of what may be expected under mild weather conditions at sources of ALAN that are typical of residential, commercial, and urban lighting in many parts of the world.

We hypothesize two different possibilities for behavioral modifications that may lead to decreased call rates at light versus dark sites. First, it is possible that birds are altering their calling behavior in response to ALAN and are calling less frequently near light sources, resulting in a reduction in recorded call rates at light sites relative to dark. However, this explanation runs counter to previous studies that have documented increased call rates of migrants when in close proximity to sources of ALAN (Van Doren et al., 2017; Watson et al., 2016). Perhaps a more likely explanation for the increase in call rates at dark sites relative to light may be negative phototaxis by migrants near ALAN. If migrants alter flight paths to avoid the unnaturally bright areas surrounding ALAN, then birds would move away from light sites (either vertically or horizontally) and would not be recorded by microphones at these sites, reducing recorded call rates compared to dark sites. In a study using a tracking radar to analyze flight paths near a small number of low-rise lights, Cabrera-Cruz et al. (2021) found that migrants were most likely to react to the lights by increasing flight height. Similarly, Bruderer et al. (1999) reported that birds increased flight height when exposed to a bright light, and Cabrera-Cruz et al. (2019) found that birds fly at greater heights in brightly lit areas, especially in the fall. Further supporting this second hypothesis, McLaren et al. (2018) found that while avian stopover density increased at regional scales in proximity to areas with bright sources of ALAN, stopover density decreased in the areas closest

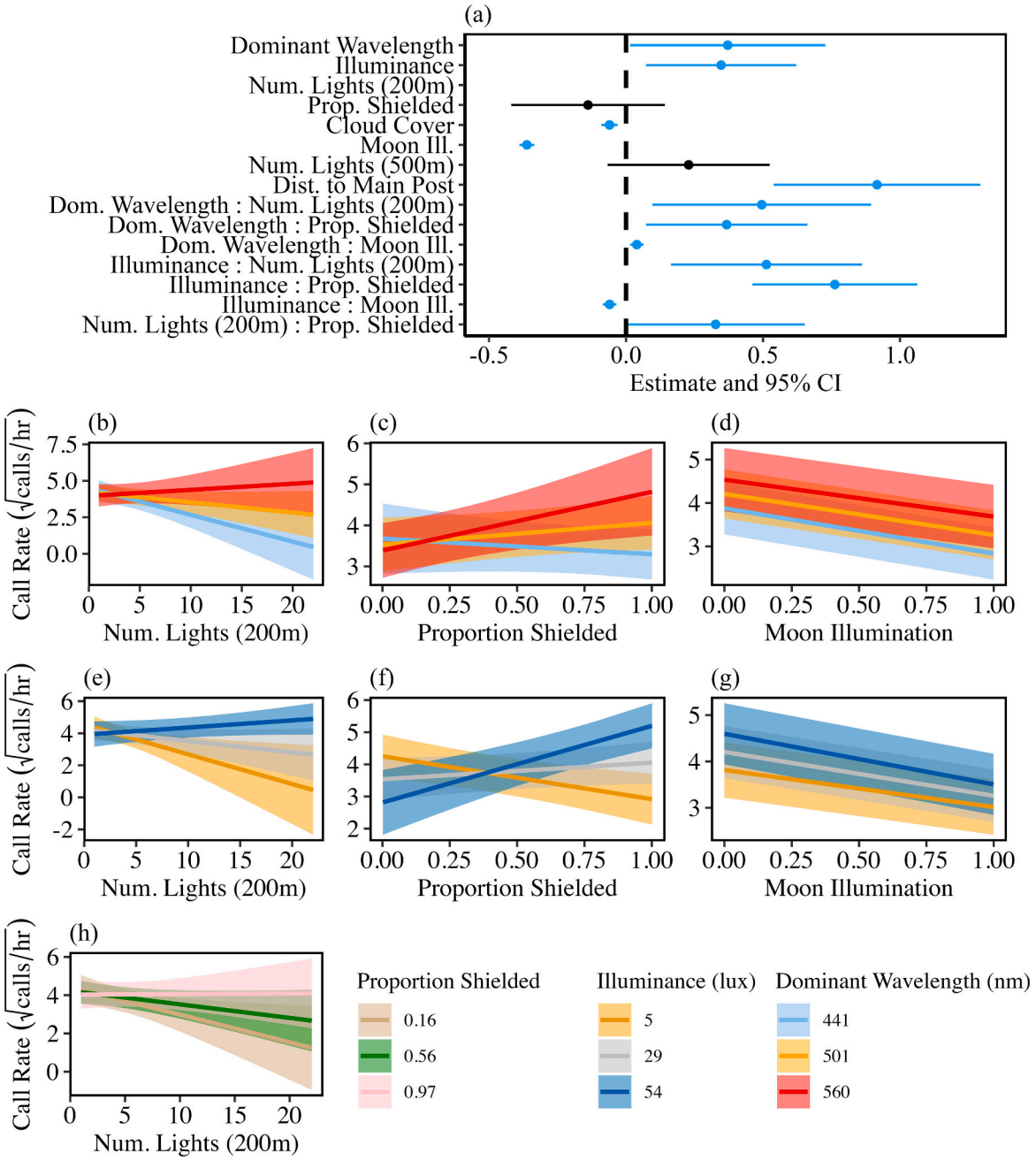


Fig. 4. Output from Model II assessing how, within lighted sites, lighting and weather variables may explain variability in call rates for data collected in southern New Mexico during fall migration of 2021, 2022, and 2023. (a) coefficient estimates and 95 % confidence intervals for each predictor in the model (dots and whiskers in black overlapped zero and are considered non-significant while those in blue do not overlap zero and are considered significant predictors of call rates), and the predicted relationships between call rates and the interactions of (b) the number of lights within 200 m of the recording location and dominant wavelength, (c) proportion shielded and dominant wavelength, (d) moon illumination and dominant wavelength, (e) the number of lights within 200 m of the recording location and illuminance, (f) proportion shielded and illuminance, (g) moon illumination and illuminance, and (h) the number of lights within 200 m of the recording location and proportion shielded. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

to the source. Therefore, it seems that at large scales, migrants are attracted to skyglow but at the local scale they avoid areas with acute sources of ALAN, except for the specific cases of poor weather conditions, extremely bright upward facing lights, or lights placed high above ground level (such as skyscrapers or radio towers). While both hypotheses presented here are possible, the data we collected do not immediately allow for testing of these alternative explanations as nocturnal flight call recordings cannot provide migrant paths or flight height information. Lastly, our finding of increased call rates at lighted sites that were located further from the Main Post (i.e., further from the most densely lit region of our study area) may indicate that the presence of sky glow may play an important role in behavioral alteration. Further work using other monitoring methods such as radar, thermal cameras, multi-dimensional nocturnal flight call recording arrays, or on-board GPS transmitters paired with accelerometers and biologgers could determine more precisely how migrating birds are behaving around light sites.

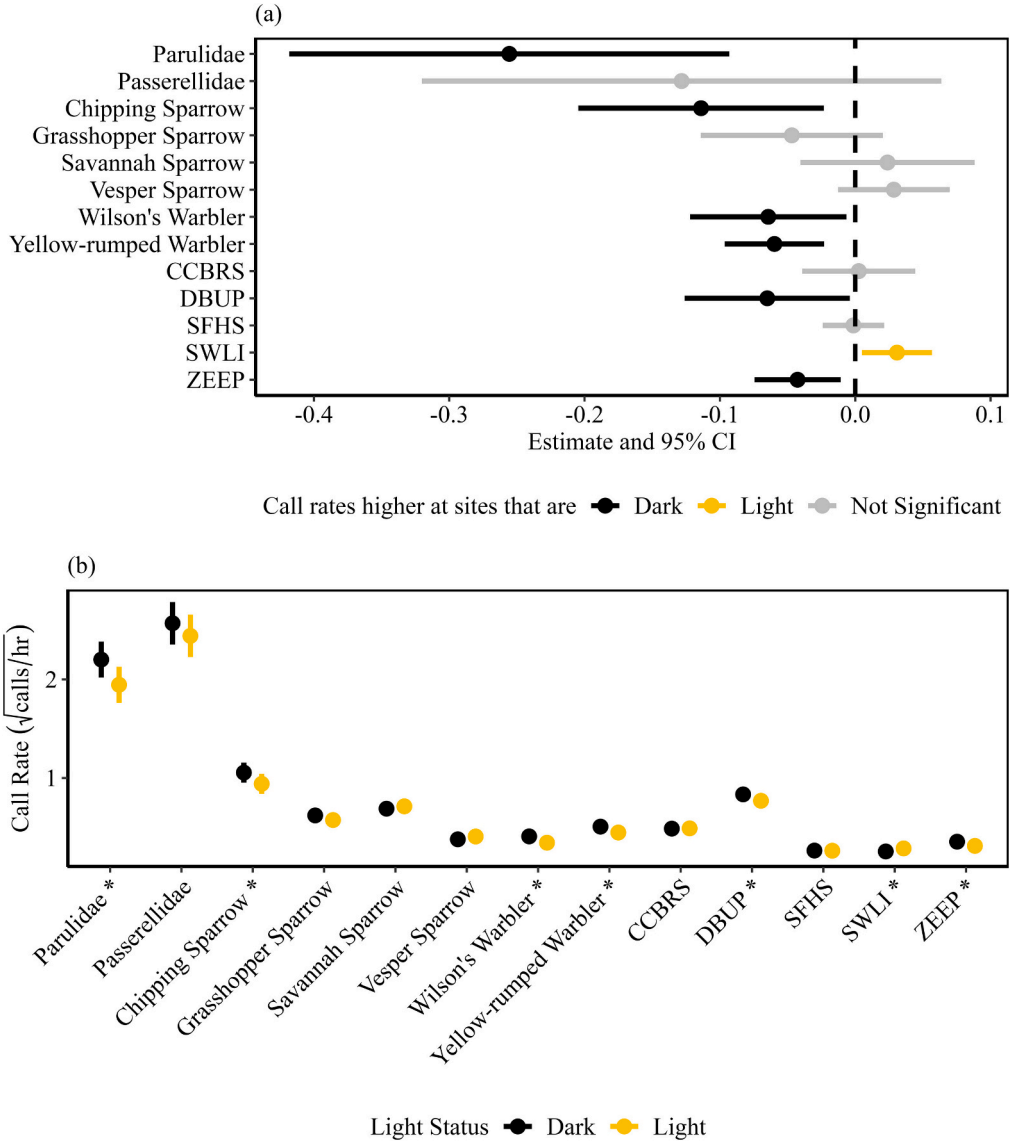


Fig. 5. (a) Coefficient estimates and 95 % confidence interval for the light status term for each of the taxanomic group models and (b) predicted relationship between call rates and light status for each of the taxonomic groups analyzed (asterisks (*) in the group name identify significant differences between light and dark sites). Definitions of groups and the species included within them can be found in Supplementary Table S1 and full model output can be found in Supplementary Table S6. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.2. Temporal variability in behavioral alteration

Across all data collection seasons, call rates were higher at dark sites than light sites, but this difference only occurred early in the migration season and disappeared entirely around the beginning of October. Differences in call rate between light and dark sites may be more distinct on nights with higher call rates, which typically occur earlier in the fall migration season when greater densities of birds are passing through our study region and night lengths are shorter (Osterhaus et al., 2024). However, the date at which call rates between light and dark sites became similar was variable—for instance, in 2023 call rates became similar around September 7th approximately 10 days before the peak call rate, and in 2022 call rates became similar around September 27th when call rates were at their peak.

While the community of migrants varies in taxonomic composition throughout the migration season in our study area, we observed relatively consistent responses across taxonomic groups that responded to ALAN, so a taxonomy-dependent response to ALAN does not appear to explain the observed difference in call rates between light and dark sites. The propensity of migrants to alter behavior in response to ALAN has been linked to age and migratory experience. In passerines, juvenile birds oriented toward simulated sky-glow in caged experiments while adults oriented in a seasonally appropriate migratory direction (Beacham, 1982; Gauthreaux, 1982; Williams, 1978). Also, in seabirds (Procellariiformes), juveniles are more likely to exhibit positive phototaxis in response to ALAN than adults, with recently fledged birds showing the highest propensity for behavioral alteration (Raine et al., 2024). Additionally, other studies have found stronger effects of ALAN in the fall, when naïve birds on their first migration represent a greater proportion of migrants, than in the spring (Cabrera-Cruz et al., 2021, 2019). Therefore, a potential explanation for the variability in response to ALAN throughout fall migration we detected is that juvenile birds early in the migration season respond most strongly to ALAN, with the strength of this response declining as birds age, complete development, become accustomed to novel stimuli such as ALAN, or are lost from the population due to mortality resulting from behavioral alteration due to ALAN. Further research into how the response of avian migrants to ALAN changes with age, development, or experience throughout the

individual's first migration season is needed.

4.3. The role of lighting variables

Shielding of lights and the use of lighting with higher dominant wavelengths ("warmer" red/yellow colored lighting) is thought to reduce the impact that these lights have on avian migrants and are recommended in "bird-friendly" lighting strategies (Burt et al., 2023a). Our finding that sites with more shielding and higher dominant wavelengths had higher call rates, which were more similar to dark sites (Fig. 4b–e), supports these strategies as ways to reduce lighting impacts to birds. Even as the number of lights within 200 m of the recording location increased, sites with higher dominant wavelength lights, and sites with more shielded lights, maintained relatively consistent call rates while call rates were reduced at other sites.

Given our finding that call rates were higher at dark sites relative to light sites, we would expect that call rates at light sites would be higher at sites with lower illuminance as these sites would more closely resemble dark sites. Contrary to this expectation, we found that as the number of lights within 200 m of the recording location increased, sites with higher illuminance had higher call rates. This discrepancy is likely due to our ground-level, human-oriented lighting measurements, which undoubtedly differs from an avian migrant's aerial perspective, and the interaction of ground-level illuminance and shielding. Shielded lights concentrate more light downward, increasing ground-level illuminance while reducing upward lighting. As a result, sites with more shielding would appear darker to birds despite higher measured illuminance at ground-level. Therefore, our finding of increased call rates at sites with greater ground-level illuminance and higher numbers of lights likely reflect avian responses to increases in shielding and decreases in upward lighting, and not the ground-level illuminance of the site itself. Further evidence for this finding can be found in the interaction between proportion shielding and illuminance, where the highest call rates occurred at sites with high illuminance and high levels of shielding, and lowest call rates occurred at sites with high illuminance and low levels of shielding. Quantifying lighting from a migrant's aerial perspective (e.g., via balloons or drones; Bobkowska et al., 2024) may be an important improvement in the design of futures studies evaluating impacts of ALAN on bird flight behavior and may provide further resolution as to how lighting variables might contribute to behavioral alteration in avian migrants.

4.4. Taxonomic variability in response

Our analysis of differences in call rates between light and dark sites for a variety of taxonomic groups revealed increases in call rates at dark sites for most groups with significant differences between site types. Interestingly, all five of the species or groups that included parulid warblers displayed an increase in call rates at dark sites in comparison to light sites. In contrast, of the eight species or groups that included passerellids, only one (Chipping Sparrow) showed higher call rates at dark sites. One (SWLI group) showed higher call rates at light sites, and the other six displayed no significant difference.

While our analyses may indicate taxonomic variability in behavioral responses to ALAN, other studies have demonstrated that both parulids and passerellids are susceptible to collisions at illuminated structures (Winger et al., 2019). The findings of Winger et al. (2019) may provide support for the hypothesis that birds are altering their calling rate when near sources of ALAN, rather than altering their flight path, as a decrease in calling rate by parulids near sources of ALAN and a maintenance of calling rate between light and dark sites by passerellids could explain the trend seen here, while still allowing for collisions. However, as mentioned above, a decrease in calling rate near sources of ALAN is counter to what has been found by other studies (Gillings and Scott, 2021; Watson et al., 2016), so further work is needed to increase our understanding of taxonomic variability in response to ALAN.

4.5. Conservation implications

While direct mortalities at illuminated structures, as seen previously on WSMR and many other locations, are a conspicuous impact of ALAN on migrating birds, our findings suggest its effects extend beyond direct mortality. Each non-fatal behavioral alteration in response to ALAN may result in excess energy expenditure, which, while possibly negligible individually, can accumulate over a migration season and result in decreased survival or fitness as birds may be exposed to hundreds or thousands of light sources along their journeys (Nemes et al., 2023). Excess energy expenditure may necessitate more frequent stopover and refueling (Lindstrom, 2003) which would prolong migration, a period in which birds suffer from high rates of mortality (Newton, 2025). Further compounding the issue, if birds must stopover in sub-optimal habitats due to energetic losses or attraction to ALAN, the recouping of energetic stores may be further delayed or even impossible.

Our findings indicate that even on nights with favorable weather conditions, avian migrants are altering their natural migration behavior in response to ALAN. ALAN, like that at WSMR, is ubiquitous in the developed world and may represent a significant hazard for nocturnal migrants. While this may be true, our findings also indicate that current "bird-friendly" lighting strategies, specifically, shielding lights and using lights of a higher dominant wavelength, may be successful in limiting behavioral alteration and thus the negative impacts of night-time lighting on avian migrants. However, while these strategies may be "bird-friendly", they still may be harmful to other taxa (including humans) that are negatively impacted by ALAN (Alaasam et al., 2021). For instance, lights with higher dominant wavelengths have been demonstrated to be more disruptive to migratory bats than lights with shorter wavelengths (Voigt et al., 2018). Also, ALAN has been demonstrated to have negative implications for human health via disruption of the circadian rhythm (Cho et al., 2015). Therefore, focusing on efforts to increase the shielding of lights, reducing the number of lights on the landscape in general, or using controls such as timers, dimmers, or motion sensors so that lights are only on when needed may be most effective in protecting the diversity of taxa impacted by ALAN.

Recognition of the negative impacts of ALAN within the environment, and the need for improved lighting policies and regulations is growing (Burt et al., 2023b; Falcón et al., 2020; Sordello et al., 2022; Yakushina, 2025). However, many current regulations surrounding light pollution and ALAN vary widely and often do not consider ecological impacts (Yakushina, 2025). As lighting policies and regulations are continuously being developed and revised, our work presented here can play an important role in improving these policies and regulations to protect nocturnal avian migrants. Specifically, our identification of specific lighting characteristics and time periods when ALAN is most impactful to migrants can aid in the creation of mitigation strategies and best practices to protect migrants from lights (Burt et al., 2023a, 2023b; Yakushina, 2022). Our work comes at a critical time when avian populations are declining widely (Johnston et al., 2025; Rosenberg et al., 2019), and the recognized importance of ecologically friendly lighting policies and regulations is increasing.

Unlike many pollutants, ALAN can be eliminated entirely simply by turning off lights. Van Doren et al. (2017) demonstrated this when avian migrants returned to normal behavior after an extremely bright ALAN source was turned off. While eliminating all ALAN is impractical, our study suggests that emphasizing the need for ecologically friendly lighting practices in future lighting policies and regulations could be an effective measure to conserve migrants and the diverse taxa negatively impacted by ALAN.

CRediT authorship contribution statement

Dylan M. Osterhaus: Writing – review & editing, Writing – original draft, Visualization, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation,

Conceptualization. **Kelley C. Boland:** Writing – review & editing. **Abigail J. Lawson:** Writing – review & editing, Methodology, Funding acquisition, Formal analysis. **Kyle G. Horton:** Writing – review & editing, Methodology, Data curation. **Benjamin M. Van Doren:** Writing – review & editing, Methodology, Funding acquisition, Formal analysis. **Patricia L. Cutler:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Timothy F. Wright:** Writing – review & editing, Supervision, Resources, Methodology, Investigation, Funding acquisition, Conceptualization. **Martha J. Desmond:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare no conflicts of interest.

Data availability

All data and code needed to recreate the analyses found herein is provided by Osterhaus et al. (2025a).

Acknowledgements

We thank T. Huycke, D. Jimenez, and F. Quiroz for their work in acoustic data collection efforts for this study. We are grateful to the editors and two reviewers who provided feedback on our manuscript. We also thank Anna Chalfoun for her review of our manuscript. We appreciate the feedback of D. Tipton and D. Eccleston on ideas presented within this manuscript and thank them for their contributions. We thank White Sands Missile Range Meteorology Branch for providing meteorological data used within this study. Our study was funded by White Sands Missile Range and financially supported by the Avian Migration Program at New Mexico State University, which is funded by USDA NIFA's HSI program (award # 2022-77040-37638), T&E Inc., Mesilla Valley Audubon Society, the New Mexico Ornithological Society, and Amazon Cloud Credits for Research Award to B. M. Van Doren. This manuscript was approved by White Sands Missile Range on 26 February 2025 for public release and unlimited distribution. Any use, trade, firm, or product names is for descriptive purposes only and does not imply endorsement by the U.S. Government.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.biocon.2025.111441.

References

Alaasam, V.J., Kernbach, M.E., Miller, C.R., Ferguson, S.M., 2021. The diversity of photosensitivity and its implications for light pollution. Integr. Comp. Biol. 61, 1170–1181. https://doi.org/10.1093/icb/icab156.

Bairlein, F., 2016. Migratory birds under threat. Science 354, 547–548. https://doi.org/10.1126/science.aah6647.

Beacham, J.L., 1982. Phototaxis and Age Dependent Migratory Orientation in the Indigo Bunting, *Passerina cyanea*. Clemson University, Clemson, South Carolina (M.S. thesis).

Berthold, P., Gwinner, E., Sonnenschein, E., 2013. Avian Migration. Springer, Berlin Heidelberg New York.

Bivand, R., Altman, M., Anselin, L., Assunção, R., Bera, A., Berke, O., Blanchet, F.G., Carvalho, M., Christensen, B., Chun, Y., Dormann, C., Dray, S., Dunnington, D., Gómez-Rubio, V., Koley, M., Kossowski, T., Krainski, E., Legendre, P., Lewin-Koh, N., Li, A., Millo, G., Mueller, W., Ono, H., Parry, J., Peres-Neto, P., Pietrzak, M., Piras, G., Reder, M., Sauer, J., Tiefelsdorf, M., Westerholt, R., Wilk, J., Wolf, L., Yu, D., 2025. Spdep: Spatial Dependence: Weighting Schemes, Statistics.

Bobkowska, K., Burdziakowski, P., Tysiac, P., Pulas, M., 2024. An innovative new approach to light pollution measurement by drone. Drones 8, 504. https://doi.org/10.3390/drones8090504.

Boland, K., 2024. An Evaluation of Avian Mortality on White Sands Missile Range and Carcass Survey Techniques. New Mexico State University, Las Cruces, New Mexico (Masters Thesis).

Brooks, M.E., Kristensen, K., Benthem, K.J., van Magnusson, A., Berg, C.W., Nielsen, A., Skaug, H.J., Mächler, M., Bolker, B.M., 2017. glmmTMB balances speed and flexibility among packages for zero-inflated generalized linear mixed modeling. R J. 9, 378. https://doi.org/10.32614/RJ-2017-066.

Bruderer, B., Peter, D., Steuri, T., 1999. Behaviour of migrating birds exposed to X-band radar and a bright light beam. J. Exp. Biol. 202, 1015–1022. https://doi.org/10.1242/jeb.202.9.1015.

Burt, C.S., Kelly, J.F., Fox, A.S., Jenkins-Smith, H.C., Leon-Corwin, M., Khalighifar, A., Trankina, G.E., Silva, C.L., Horton, K.G., 2023a. Can ecological forecasting lead to convergence on sustainable lighting policies? Conserv. Sci. Pract. 5, e12920. https://doi.org/10.1111/csp2.12920.

Burt, C.S., Kelly, J.F., Trankina, G.E., Silva, C.L., Khalighifar, A., Jenkins-Smith, H.C., Fox, A.S., Fristrup, K.M., Horton, K.G., 2023b. The effects of light pollution on migratory animal behavior. Trends Ecol. Evol. 38, 355–368. https://doi.org/10.1016/j.tree.2022.12.006.

Cabrera-Cruz, S.A., Smolinsky, J.A., Buler, J.J., 2018. Light pollution is greatest within migration passage areas for nocturnally-migrating birds around the world. Sci. Rep. 8, 3261. https://doi.org/10.1038/s41598-018-21577-6.

Cabrera-Cruz, S.A., Smolinsky, J.A., McCarthy, K.P., Buler, J.J., 2019. Urban areas affect flight altitudes of nocturnally migrating birds. J. Anim. Ecol. 88, 1873–1887. https://doi.org/10.1111/1365-2656.13075.

Cabrera-Cruz, S.A., Cohen, E.B., Smolinsky, J.A., Buler, J.J., 2020. Artificial light at night is related to broad-scale stopover distributions of nocturnally migrating landbirds along the Yucatan peninsula, Mexico. Remote Sens. 12, 395. https://doi.org/10.3390/rs12030395.

Cabrera-Cruz, S.A., Larkin, R.P., Gimpel, M.E., Gruber, J.G., Zenzal, T.J., Buler, J.J., 2021. Potential effect of low-rise, downcast artificial lights on nocturnally migrating land birds. Integr. Comp. Biol. 61, 1216–1236. https://doi.org/10.1093/icb/icab154.

Cho, Y., Ryu, S.-H., Lee, B.R., Kim, K.H., Lee, E., Choi, J., 2015. Effects of artificial light at night on human health: a literature review of observational and experimental studies applied to exposure assessment. Chronobiol. Int. 32, 1294–1310. https://doi.org/10.3109/07420528.2015.1073158.

Dokter, A.M., Desmet, P., Spaaks, J.H., Van Hoey, S., Veen, L., Verlinden, L., Nilsson, C., Haase, G., Leijnse, H., Farnsworth, A., Bouten, W., Shamoun-Baranes, J., 2019. bioRad: biological analysis and visualization of weather radar data. Ecography 42, 852–860. https://doi.org/10.1111/ecog.04028.

Evans, R., Mellinger, D.K., 1999. Monitoring grassland birds in nocturnal migration. Stud. Avian Biol. 219–229.

Evans, W.R., Rosenberg, K.V., 2000. Acoustic monitoring of night-migrating birds: a progress report. In: Acoustic Monitoring of Night-migrating Birds: A Progress Report, pp. 151–159.

Falchi, F., Cinzano, P., Duriscoe, D., Kyba, C.C.M., Elvidge, C.D., Baugh, K., Portnov, B. A., Rybnikova, N.A., Furgoni, R., 2016. The new world atlas of artificial night sky brightness. Sci. Adv. 2, e1600377. https://doi.org/10.1126/sciadv.1600377.

Falcón, J., Torriglia, A., Attia, D., Viénot, F., Gronfier, C., Behar-Cohen, F., Martinsons, C., Hicks, D., 2020. Exposure to artificial light at night and the consequences for flora, fauna, and ecosystems. Front. Neurosci. 14, 602796. https://doi.org/10.3389/fnins.2020.602796.

Gaston, K.J., Blackburn, T.M., Goldewijk, K.K., 2003. Habitat conversion and global avian biodiversity loss. Proc. R. Soc. Lond. B 270, 1293–1300. https://doi.org/10.1098/rspb.2002.2303.

Gauthreaux, S.A., 1982. Age-dependent orientation in migratory birds. In: Avian Navigation. Presented at the International Symposium on Avian Navigation. Springer, Tirrenia, Pisa, pp. 68–74.

Gauthreaux Jr., S.A., Belser, C.G., Rich, C., Longcore, T., 2006. Effects of artificial night lighting on migrating birds. In: Ecological Consequences of Artificial Night Lighting. Island Press, Washington, DC, USA, pp. 67–93.

Gillings, S., Scott, C., 2021. Nocturnal flight calling behaviour of thrushes in relation to artificial light at night. Ibis 163, 1379–1393. https://doi.org/10.1111/ibi.12955.

Horton, K.G., Shriver, W.G., Buler, J.J., 2015. A comparison of traffic estimates of nocturnal flying animals using radar, thermal imaging, and acoustic recording. Ecol. Appl. 25, 390–401. https://doi.org/10.1890/14-0279.1.

Horton, K.G., Nilsson, C., Van Doren, B.M., La Sorte, F.A., Dokter, A.M., Farnsworth, A., 2019. Bright lights in the big cities: migratory birds' exposure to artificial light. Front. Ecol. Environ. 17, 209–214. https://doi.org/10.1002/fee.2029.

Horton, K.G., Buler, J.J., Anderson, S.J., Burt, C.S., Collins, A.C., Dokter, A.M., Guo, F., Sheldon, D., Tomaszewska, M.A., Henebry, G.M., 2023. Artificial light at night is a top predictor of bird migration stopover density. Nat. Commun. 14, 7446. https://doi.org/10.1038/s41467-023-43046-z.

Hung, L.-W., Anderson, S.J., Pipkin, A., Fristrup, K., 2021. Changes in night sky brightness after a countywide LED retrofit. J. Environ. Manag. 292, 112776. https://doi.org/10.1016/j.jenvman.2021.112776.

Johnston, A., Rodewald, A.D., Strimas-Mackey, M., Auer, T., Hochachka, W.M., Stillman, A.N., Davis, C.L., Ruiz-Gutierrez, V., Dokter, A.M., Miller, E.T., Robinson, O., Ligocki, S., Jaromczyk, L.O., Crowley, C., Wood, C.L., Fink, D., 2025. North American bird declines are greatest where species are most abundant. Science 388, 532–537. https://doi.org/10.1126/science.adn4381.

Jones, T., Cresswell, W., 2010. The phenology mismatch hypothesis: are declines of migrant birds linked to uneven global climate change? J. Anim. Ecol. 79, 98–108. https://doi.org/10.1111/j.1365-2656.2009.01610.x.

Lindstrom, A., 2003. Fuel deposition rates in migrating birds: causes, constraints and consequences. In: Avian Migration. Springer, New York, NY, USA, pp. 307–320.

Loss, S.R., Will, T., Marra, P.P., 2015. Direct mortality of birds from anthropogenic causes. Annu. Rev. Ecol. Evol. Syst. 46, 99–120. https://doi.org/10.1146/annurev-ecolsys-112414-054133.

Loss, S.R., Lao, S., Eckles, J.W., Anderson, A.W., Blair, R.B., Turner, R.J., 2019. Factors influencing bird-building collisions in the downtown area of a major North American city. PLoS One 14, e0224164. https://doi.org/10.1371/journal.pone.0224164.

McLaren, J.D., Buler, J.J., Schreckengost, T., Smolinsky, J.A., Boone, M., Emiel Van Loon, E., Dawson, D.K., Walters, E.L., 2018. Artificial light at night confounds broad-scale habitat use by migrating birds. Ecol. Lett. 21, 356–364. https://doi.org/10.1111/ele.12902.

Nemes, C.E., Cabrera-Cruz, S.A., Anderson, M.J., DeGroote, L.W., DeSimone, J.G., Massa, M.L., Cohen, E.B., 2023. More than mortality: consequences of human activity on migrating birds extend beyond direct mortality. Ornithol. Appl. 125, duad020. https://doi.org/10.1093/ornithapp/duad020.

Newton, I., 2025. Migration mortality in birds. Ibis 167, 106–123. https://doi.org/10.1111/ibi.13316.

Osterhaus, D.M., Van Doren, B.M., Horton, K.G., Abadi, F., Wright, T.F., Desmond, M.J., 2024. Evaluation of methods to estimate nocturnal bird migration activity: a comparison of radar and nocturnal flight call monitoring in the American west. Ornithol. Appl., duae062 https://doi.org/10.1093/ornithapp/duae062.

Osterhaus, D.M., Boland, K.C., Lawson, A.J., Horton, K.G., Van Doren, B.M., Cutler, P.L., Wright, T.F., Desmond, M.J., 2025a. Data and code from: Nocturnal flight call monitoring reveals in-flight behavioral alteration by avian migrants in response to artificial light at night. https://doi.org/10.5061/dryad.w6m905r26.

Osterhaus, D.M., Boland, K.C., Horton, K.G., Van Doren, B.M., Cutler, P.L., Desmond, M. J., Wright, T.F., 2025b. A call in the dark: nocturnal flight calls and their potential to advance the study of avian migration. Ornithology, ukaf002. https://doi.org/10.1093/ornithology/ukaf002.

R Core Team, 2023. R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna, Austria. http://www.R-project.org/.

Raine, A.F., Driskill, S., Rothe, J., Rossiter, S., Gregg, J., Anderson, T., Travers, M.S., 2024. The impact of light attraction on adult seabirds and the effectiveness of minimization actions. Pac. Sci. 78, 85–102.

Rebke, M., Dierschke, V., Weiner, C.N., Aumüller, R., Hill, K., Hill, R., 2019. Attraction of nocturnally migrating birds to artificial light: the influence of colour, intensity and blinking mode under different cloud cover conditions. Biol. Conserv. 233, 220–227. https://doi.org/10.1016/j.biocon.2019.02.029.

Roper, D., 2015. Veterinary Pathology Examination Final Report (No. 2015204190). United States Department of the Interior: US Fish and Wildlife Service.

Rosenberg, K.V., Dokter, A.M., Blancher, P.J., Sauer, J.R., Smith, A.C., Smith, P.A., Stanton, J.C., Panjabi, A., Helft, L., Parr, M., Marra, P.P., 2019. Decline of the North American avifauna. Science 366, 120–124. https://doi.org/10.1126/science.aaw1313.

Saino, N., Ambrosini, R., Rubolini, D., Von Hardenberg, J., Provenzale, A., Hüppop, K., Hüppop, O., Lehikoinen, A., Lehikoinen, E., Rainio, K., Romano, M., Sokolov, L., 2011. Climate warming, ecological mismatch at arrival and population decline in migratory birds. Proc. R. Soc. B 278, 835–842. https://doi.org/10.1098/rspb.2010.1778.

Sordello, R., Busson, S., Cornuau, J.H., Deverchère, P., Faure, B., Guetté, A., Hölker, F., Kerbiriou, C., Lengagne, T., Le Viol, I., Longcore, T., Moeschler, P., Ranzoni, J., Ray, N., Reyjol, Y., Roulet, Y., Schroer, S., Secondi, J., Valet, N., Vanpeene, S., Vauclair, S., 2022. A plea for a worldwide development of dark infrastructure for biodiversity – practical examples and ways to go forward. Landsc. Urban Plan. 219, 104332. https://doi.org/10.1016/j.landurbplan.2021.104332.

Tan, D.J.X., Freymueller, N.A., Teo, K.M., Symes, W.S., Lum, S.K.Y., Rheindt, F.E., 2024. Disentangling the biotic and abiotic drivers of bird–building collisions in a tropical Asian city with ecological niche modeling. Conserv. Biol. 38, e14255. https://doi.org/10.1111/cobi.14255.

Thieurmel, B., Elmarhraoui, A., 2022. Suncalc: Compute Sun Position, Sunlight Phases, Moon Position and Lunar Phase.

Van Doren, B.M., Horton, K.G., Dokter, A.M., Klinck, H., Elbin, S.B., Farnsworth, A., 2017. High-intensity urban light installation dramatically alters nocturnal bird migration. Proc. Natl. Acad. Sci. USA 114, 11175–11180. https://doi.org/10.1073/pnas.1708574114.

Van Doren, B.M., Willard, D.E., Hennen, M., Horton, K.G., Stuber, E.F., Sheldon, D., Sivakumar, A.H., Wang, J., Farnsworth, A., Winger, B.M., 2021. Drivers of fatal bird collisions in an urban center. Proc. Natl. Acad. Sci. USA 118, e2101666118. https://doi.org/10.1073/pnas.2101666118.

Van Doren, B.M., Farnsworth, A., Stone, K., Osterhaus, D.M., Drucker, J., Van Horn, G., 2024. Nighthawk: acoustic monitoring of nocturnal bird migration in the Americas. Methods Ecol. Evol. 15, 329–344. https://doi.org/10.1111/2041-210X.14272.

Voigt, C.C., Rehnig, K., Lindecke, O., Pētersons, G., 2018. Migratory bats are attracted by red light but not by warm-white light: implications for the protection of nocturnal migrants. Ecol. Evol. 8, 9353–9361. https://doi.org/10.1002/ece3.4400.

Watson, M.J., Wilson, D.R., Mennill, D.J., 2016. Anthropogenic light is associated with increased vocal activity by nocturnally migrating birds. Condor 118, 338–344. https://doi.org/10.1650/CONDOR-15-136.1.

Williams, J., 1978. Age-Dependent Factors in the Migratory Behavior of the White-crowned Sparrow, *Zonotrichia leucophrys*. Clemson University, Clemson, South Carolina (MS Thesis).

Winger, B.M., Weeks, B.C., Farnsworth, A., Jones, A.W., Hennen, M., Willard, D.E., 2019. Nocturnal flight-calling behaviour predicts vulnerability to artificial light in migratory birds. Proc. R. Soc. B 286, 20190364. https://doi.org/10.1098/rspb.2019.0364.

Yakushina, Y., 2022. The response of the Bonn convention on migratory species to light pollution. IOP Conf. Ser. Earth Environ. Sci. 1099, 012048. https://doi.org/10.1088/1755-1315/1099/1/012048.

Yakushina, Y., 2025. Light pollution regulations and where to find them. J. Environ. Manag. 373, 123757. https://doi.org/10.1016/j.jenvman.2024.123757.

Zhao, X., Chen, M., Wu, Z., Wang, Z., 2014. Factors influencing phototaxis in nocturnal migrating birds. Zool. Sci. 31, 781–788. https://doi.org/10.2108/zs130237.

Zhao, X., Zhang, M., Che, X., Zou, F., 2020. Blue light attracts nocturnally migrating birds. Condor 122, duaa002. https://doi.org/10.1093/condor/duaa002.